Submitted to Statistical Science

A Sample Article Title

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Abstract. The abstract should summarize the contents of the paper. It should be clear, descriptive, self-explanatory and not longer than 200 words. It should also be suitable for publication in abstracting services. Formulas should be used as sparingly as possible within the abstract. The abstract should not make reference to results, bibliography or formulas in the body of the paper—it should be self-contained.

Key words and phrases: first keyword, second keyword

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1. INTRODUCTION

This template helps you to create a properly formatted manuscript for IMS journals. Set the `journal` field in the YAML header to the appropriate journal code (e.g., `sts` for *Statistical Science*, `aos` for *Annals of Statistics*). Prepare your paper in the same style as used in this sample. Try to avoid excessive use of italics and bold face.

1.1 Citation

For numbered citation journals, always use square brackets. For example, [1] or for multiple bibliography items [1–4]. For name-year citation journals, you can drop the square brackets for in-text citations.

2. FONTS

Please use text fonts in text mode, e.g.:

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- **Bold**
- SMALL CAPS
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Please use mathematical fonts in mathematical mode, e.g.:

- $\mathrm{ABCabc123}$
- $ABCabc123$
- $\mathbf{ABCabc123}$
- $\boldsymbol{ABCabc123\alpha\beta\gamma}$
- $\mathcal{ABC}$
- $\mathbb{ABC}$
- $\mathsf{ABCabc123}$
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- $\mathfrak{ABCabc123}$

Note that `\mathcal` and `\mathbb` belong to capital letters-only font typefaces.

Name1 Surname1, Department, University or Company Name, City, Country (e-mail: first@somewhere.com). Name2 Surname2, Department, Another University, City, Country (e-mail: second@somewhere.com). Name3 Surname3, Department, Third University, City, Country (e-mail: third@somewhere.com).

[1]Some comment

3. NOTES

Footnotes[1] pose no problem.[2]

4. ENVIRONMENTS

4.1 Examples for Plain-Style Environments

Theorem environments are defined in `header-includes` and used as raw LaTeX blocks. This is required because the IMS class patches `\theoremstyle` in ways that are incompatible with Quarto's automatic theorem injection.

AXIOM 1. *This is the body of Axiom 1.*

PROOF. This is the body of the proof of the axiom above. □

CLAIM 2. *This is the body of Claim 2. Claim 2 is numbered after Axiom 1 because we used* `[axiom]` *in* `\newtheorem`.

THEOREM 4.1. *This is the body of Theorem 4.1. Theorem 4.1 numbering is dependent on section because we used* `[section]` *after* `\newtheorem`.

THEOREM 4.2 (Title of the theorem). *This is the body of Theorem 4.2. Theorem 4.2 has additional title.*

LEMMA 4.3. *This is the body of Lemma 4.3. Lemma 4.3 is numbered after Theorem 4.2 because we used* `[theorem]` *in* `\newtheorem`.

PROOF OF LEMMA 4.3. This is the body of the proof of Lemma 4.3. □

4.2 Examples for Definition-Style Environments

DEFINITION 4.4. This is the body of Definition 4.4. Definition 4.4 is numbered after Lemma 4.3 because we used `[theorem]` in `\newtheorem`.

EXAMPLE. This is the body of the example. Example is unnumbered because we used `\newtheorem*` instead of `\newtheorem`.

FACT. This is the body of the fact. Fact is unnumbered because we used `\newtheorem*` instead of `\newtheorem`.

5. TABLES AND FIGURES

Sample of cross-reference to figure. Figure 1 shows the estimated density of the standard normal distribution.

Cross-references to labeled tables: see Table 1.

6. EQUATIONS AND THE LIKE

Two equations:

$$C_s = K_M \frac{\mu/\mu_x}{1-\mu/\mu_x} \tag{1}$$

[1] This is an example of a footnote.

[2] Note that footnote number is after punctuation.

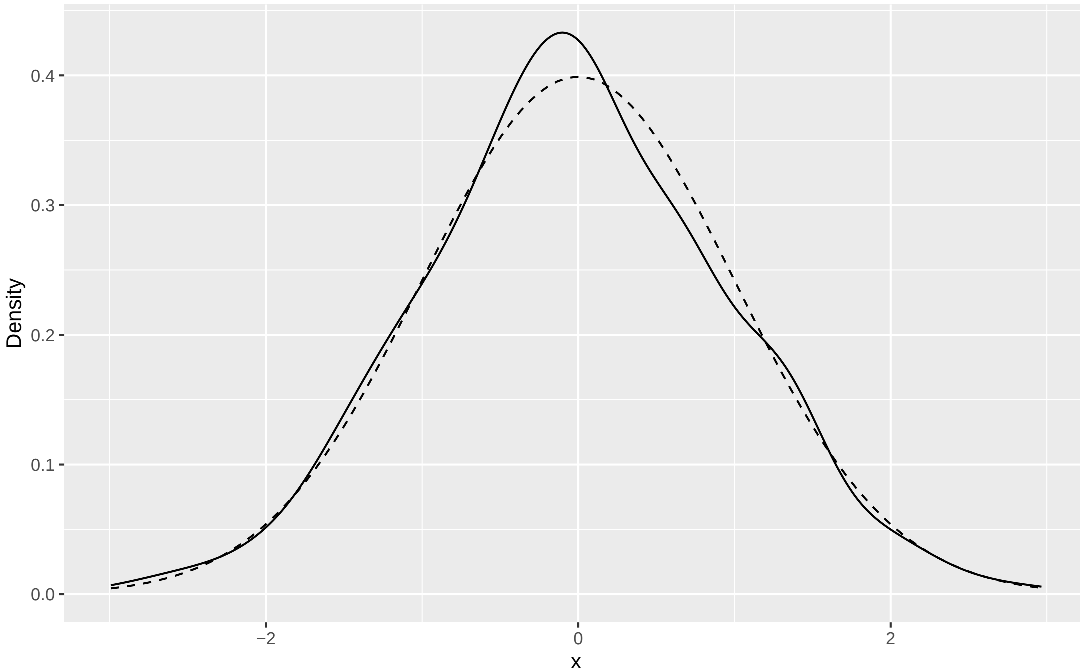

Figure 1: Kernel density estimate of 500 standard normal samples.

TABLE 1
Linear model coefficient estimates (mtcars)

Parameter	Mean	Std. dev	2.5%	50%	97.5%
(Intercept)	37.227	1.599	33.957	37.227	40.497
wt	-3.878	0.633	-5.172	-3.878	-2.584
hp	-0.032	0.009	-0.050	-0.032	-0.013

and

$$G = \frac{P_{\text{opt}} - P_{\text{ref}}}{P_{\text{ref}}} 100(\%).$$

Aligned equations:

$$\frac{dS}{dt} = -\sigma X + s_F F, \tag{2}$$

$$\frac{dX}{dt} = \mu X, \tag{3}$$

$$\frac{dP}{dt} = \pi X - k_h P, \tag{4}$$

$$\frac{dV}{dt} = F. \tag{5}$$

One long equation:

$$\begin{aligned}\mu_{\text{normal}} &= \mu_x \frac{C_s}{K_x C_x + C_s}\\ &= \mu_{\text{normal}} - Y_{x/s}\big(1 - H(C_s)\big)(m_s + \pi/Y_{p/s})\\ &= \mu_{\text{normal}}/Y_{x/s} + H(C_s)(m_s + \pi/Y_{p/s}).\end{aligned}$$

7. APPENDIX

Appendices should be provided in the Appendix section, before Acknowledgements. If there is only one appendix, please refer to it as the Appendix. If there are multiple appendices, refer to them as Appendix A, Appendix B, etc.

ACKNOWLEDGMENTS

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SUPPLEMENTARY MATERIAL

Title of Supplement A
Short description of Supplement A.

Title of Supplement B
Short description of Supplement B.

REFERENCES

[1] BILLINGSLEY, P. (1999). *Convergence of Probability Measures*, 2 ed. Wiley, New York.
[2] BOURBAKI, N. (1966). *General Topology* **1**. Addison-Wesley, Reading, MA.
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